Dual Monte Carlo uncertainty framework for neural networks: application to a quantum voltage standard

Guillermo Schneider – Instituto Nacional de Tecnología Industrial (INTI), INCALIN-UNSAM, Argentina
Ricardo Iuzzolino – Instituto Nacional de Tecnología Industrial (INTI), Argentina
Mariano Real – Instituto Nacional de Tecnología Industrial (INTI), INCALIN-UNSAM, Argentina
Mehedin Arifovic – TÜBITAK Ulusal Metroloji Enstitüsü, Türkiye
Giacomo Maculotti – DIGEP, Politecnico di Torino, Italy
Guillermo Schneider – gschneider@inti.gob.ar

INTRODUCTION

Programmable Josephson Voltage Standards (PJVS) [1] enable traceable DC voltage realization with extremely low uncertainty. Neural-network-based digital twins can model PJVS behavior for prediction and optimization, but their uncertainty contributions must be quantified in compliance with the GUM [2] for metrological use. This work proposes the Dual Monte Carlo Uncertainty Framework (DMCUF), which combines GUM-compliant Monte Carlo [3] propagation of measurement uncertainties with Monte Carlo Dropout [4] —to estimate model uncertainty— and bias uncertainty evaluation from reference measurements. The method is demonstrated using DC voltage measurements from a PJVS, part of a neural-network-based digital twin developed within the scope of the EURAMET ViDiT project [5] to optimize the system and reduce setup and measurement times.

METHOD

The datasets were generated using measurements of a DC voltage step of 1.0001086679 V, from the TÜBITAK UME 10 V PJVS system, with the following experimental setup:

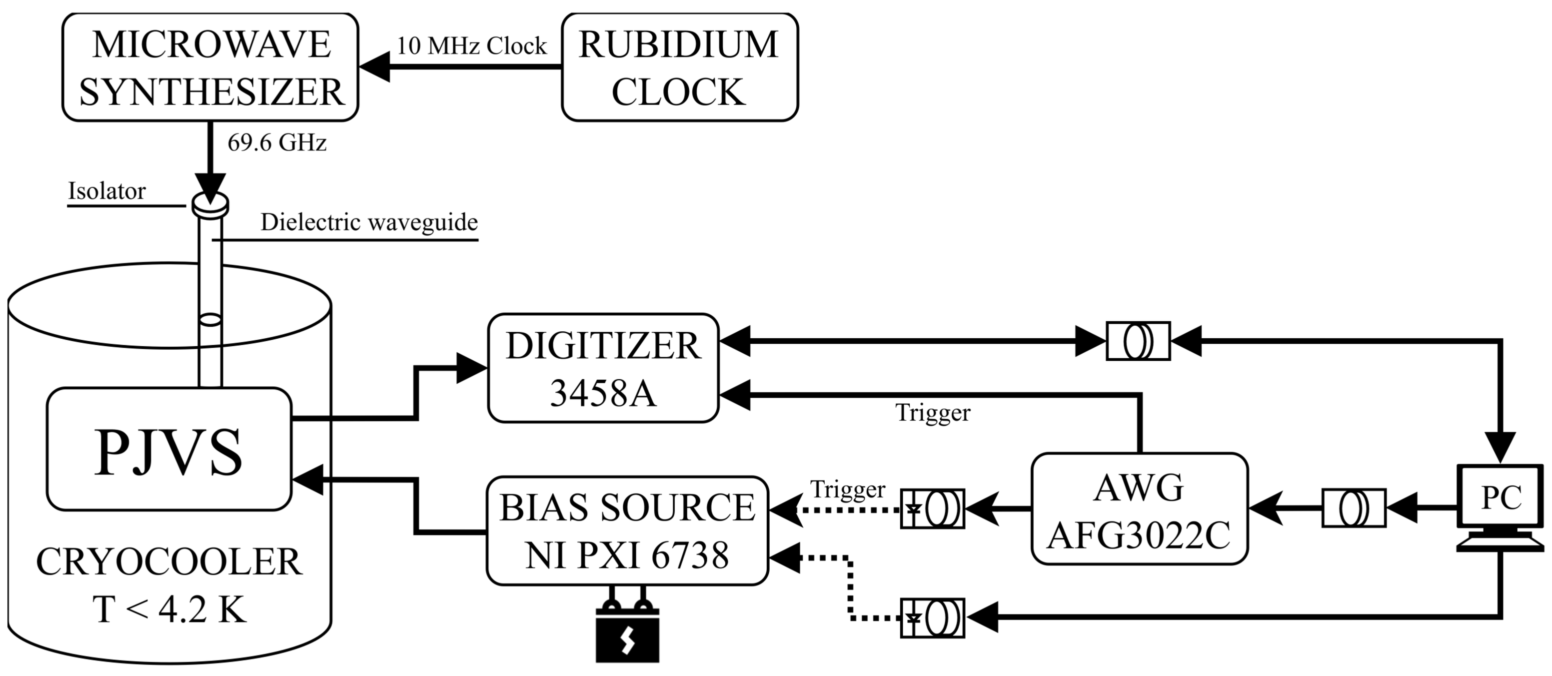

Fig. 1 – Experimental setup

A neural network is trained on the system's measurements, and the bias of the model's predictions is obtained as the mean of the residuals between reference measurements and the predictions, from which we can obtain the model bias uncertainty. The total variance is decomposed using the law of total variance.

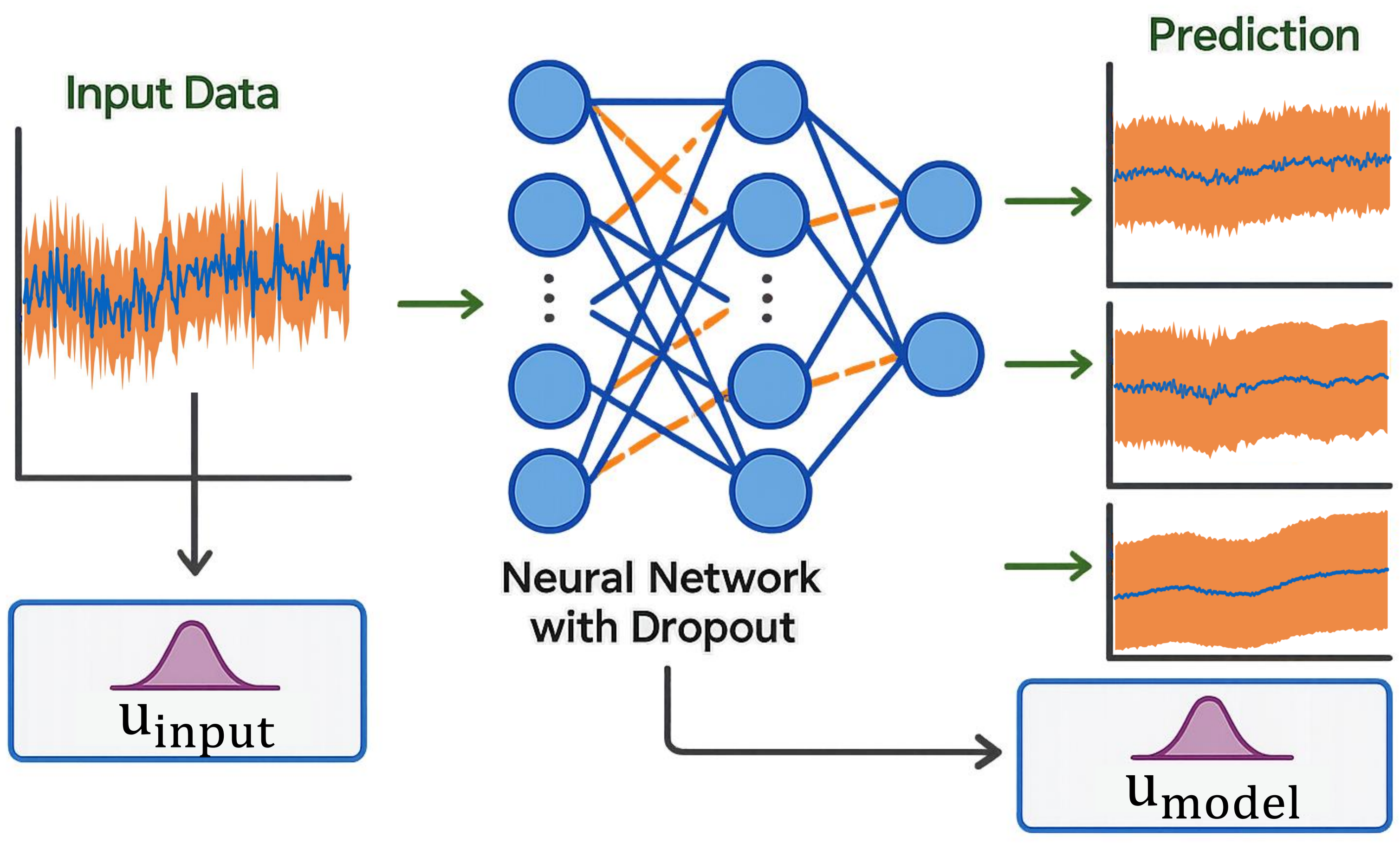

Fig. 2 – Combination of Monte Carlo and Monte Carlo Dropout methods

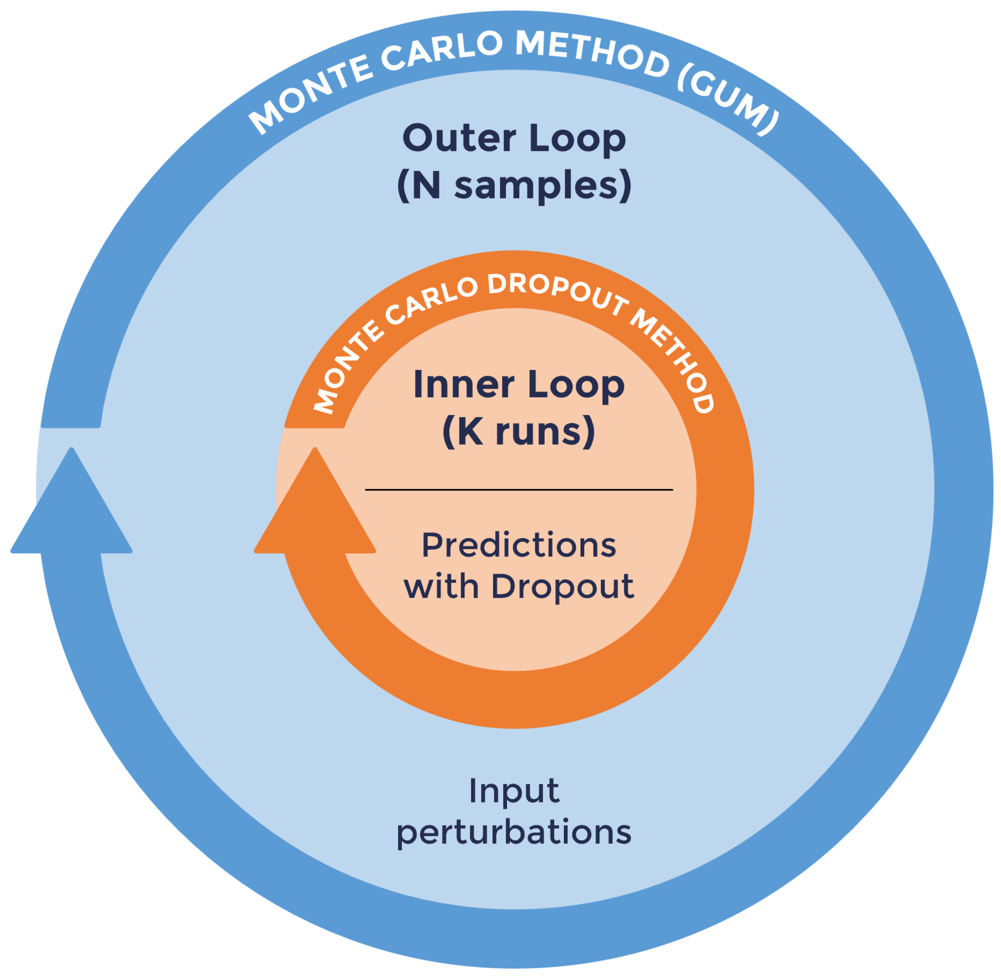

Fig. 3 – Representation of the double loop uncertainty evaluation method

RESULTS

Three architectures were tested as surrogate models of the PJVS digital twin: a multilayer perceptron (MLP), a long short-term memory network (LSTM), and a convolutional neural network (CNN). All models were trained separating in 70% training, 20% validation, and 10% test in a one-step-ahead sequence-to-sequence setting. The DMCUF used N = 1000 Monte Carlo input samples with an uncertainty of 79 nV estimated from the measurement series, and K = 200 Monte Carlo Dropout realizations (dropout rate = 40%). Coverage factor is k = 1.

Table 1 - Comparison of predictive performance (test set)

Model	RMSE (nV)	Bias (nV)	Coverage 95%
MLP	**64**	**-2**	**96.47%**
LSTM	66	-13	90.93%
CNN	69	-17	73.05%

Table 2 - Comparison of uncertainty components (test set)

Model	u_b (nV)	$\langle u_{input} \rangle$ (nV)	$\langle u_{model} \rangle$ (nV)	$\langle u_c \rangle$ (nV)
MLP	**3**	**22**	**67**	**71**
LSTM	3	16	50	53
CNN	3	9	33	35

CONCLUSIONS

DMCUF provides a GUM-compliant uncertainty evaluation method for neural-network-based models, as demonstrated in this PJVS digital twin application. It provides a transparent and reproducible way to quantify uncertainty contributions from both inputs and model structure. Results showed that combined uncertainty is strongly dependent on model architecture and dropout layer configuration. Due to its superior predictive performance, the MLP model proved to be the one that best represents the measured data. The approach presented is easily applicable to other metrological systems that employ data-driven models.

REFERENCES

[1] S. Bauer et al. Measurement Science and Technology, 2022.
[2] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, OIML. GUM, JCGM 100:2008.
[3] BIPM, IEC, IFCC, ILAC, ISO, IUPAC, IUPAP, OIML. GUM Sup. 1, JCGM 101:2008.
[4] Y. Gal and Z. Ghahramani. 33rd International Conference on Machine Learning, 2016.
[5] ViDiT Project, "22dit01 vidit: Trustworthy virtual experiments and digital twins, 2024.

This work was financed partially by 22DIT01 ViDiT Project. The project (22DIT01 ViDiT) has received funding from the European Partnership on Metrology, co-financed from the European Union's Horizon Europe Research and Innovation Programme and by the Participating States.